Math Primer
(19. Named Distributions)

Contents

C.F. Gauss
(1777-1855)

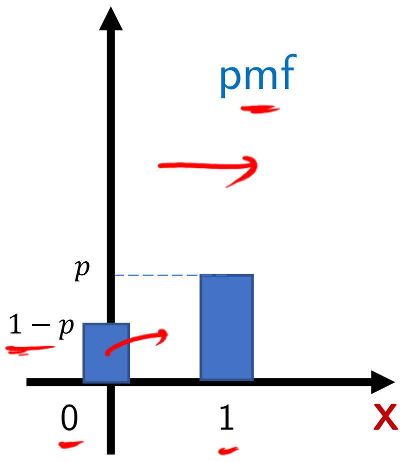
(discrete distributions)

Bernoulli Distribution

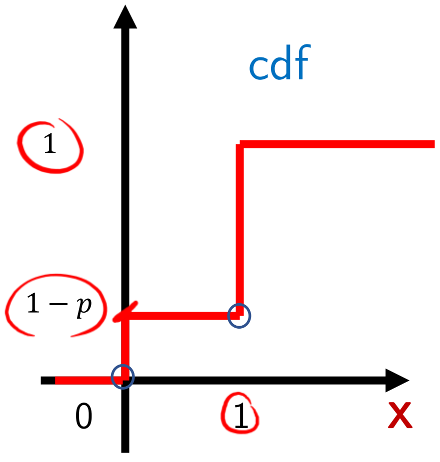
- Used to model a phenomenon with binary outcomes (dichotomous) : True/False, Success/Failure, Default/No Default etc.

prob. of success

Parameters	$p \in (0,1)$
Support	$x \in \{0, 1\}$
PMF	$\begin{cases} 1-p, & x = 0 \\ p, & x = 1 \end{cases}$ failure / success
PMF (alternate)	$p^x(1-p)^{1-x}$ x=1, p; x=0, 1-p
Mean	p
Variance	$p(1-p)$



$$P_X(x) = p^x(1-p)^{1-x}$$



$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

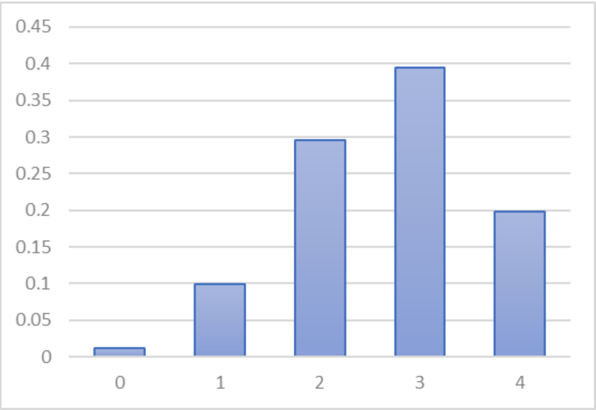
Binomial Distribution

- No. of successes in 'n' *independent* and *identical* Bernoulli trials

$X = ?$ # success n, p

Parameters	$n \in \mathbb{N}, p \in (0,1)$
Support	$x \in \{0, 1, 2, \ldots, n\}$ (1, 2..)
PMF P(X=x)	$\binom{n}{x} p^x (1-p)^{n-x}$
CDF	$\sum_{x=0}^{x=x} \binom{n}{x} p^x (1-p)^{n-x}$
Mean	np
Variance	$np(1-p)$

pmf $(n = 4, p = 0.6)$



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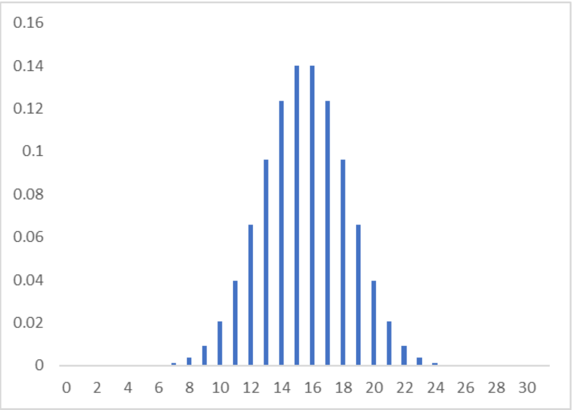
pmf $(n = 30, p = 0.5)$



Questions

PEAKS2TAILS

Exercise 1 – A student as 60% chances of answering a question correctly. What is the probability that

a) The student answers exactly 2 out of 4 questions correctly

b) The student answers at least 2 out of 4 questions correctly

$X = \#$ correct answers, binomial

$X \sim \text{Binom}(n=4, p=0.6)$

a) $\mathbb{P}(X=2) = ({}^4C_2)(0.6)^2 \cdot (0.4)^2 = 0.6912$

b) $\mathbb{P}(X \geq 2) = \mathbb{P}(X=2) + \mathbb{P}(X=3) + \mathbb{P}(X=4)$

Exercise 2 – A bank has a homogeneous loan portfolio of 5 loans. Each loan amount is \$ 100. Each borrower has a probability of default (PD) = 5%. Assuming independence among default events and no recovery, find

a) Expected Loss

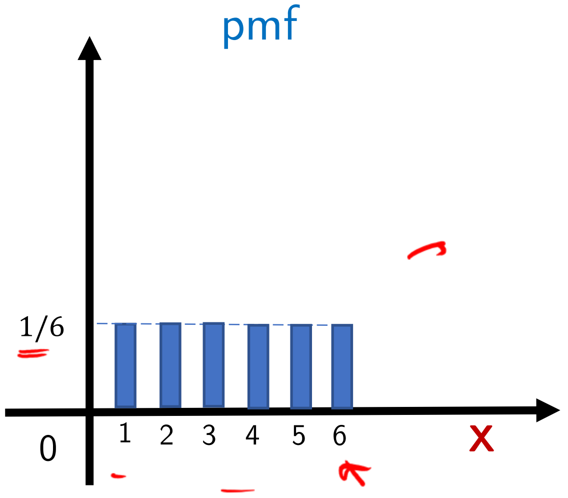
b) A conservative loss figure that the bank will exceed only 5% of the time (=won't exceed 95% of the time) VaR(95%)

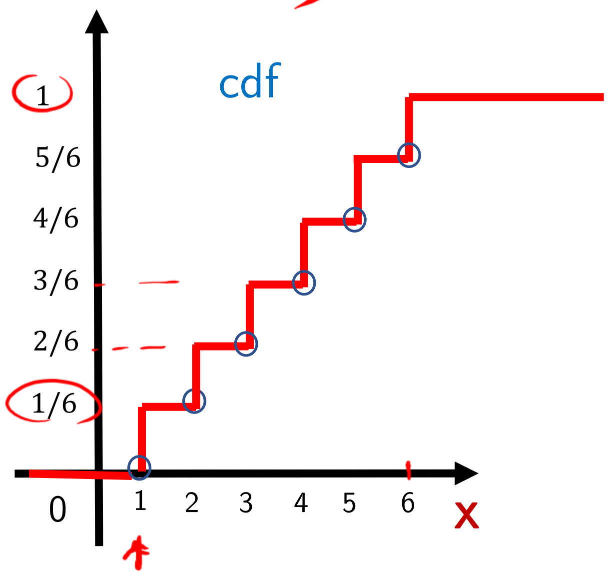
Uniform Distribution

PEAKS2TAILS

- Used to model a phenomenon with equally likely outcomes : (rolling a die)

Parameters	a, b
Support	$x \in \{a, a+1, \ldots, b-1, b\}$ $n = b - a + 1$
PMF	$1/n$
CDF $x = a$ → $1/n$ $(a+0.5)$ $a+1$ $(2/n)$	$\dfrac{\lfloor x \rfloor - a + 1}{n}$
Mean	$\dfrac{a+b}{2}$
Variance	$\dfrac{n^2 - 1}{12}$





$\lfloor 3 \rfloor = 3$

$\lfloor 3.5 \rfloor = 3$

Poisson Distribution

- Used to model number of occurrences of a particular event per unit
  - Customers arriving to a mall on a day ($\lambda = 100$ customers per hr.)
  - Number of defaults in a portfolio of 1 million obligors

λ

np

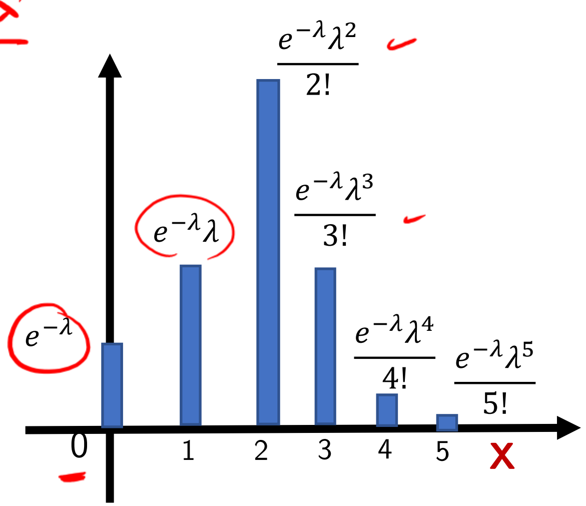
$\frac{e^{-\lambda} \cdot \lambda^0}{0!} = e^{-\lambda}$

$\frac{e^{-\lambda} \cdot \lambda^1}{1!}$

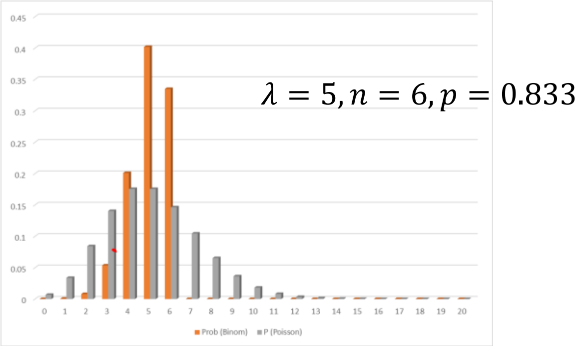
Parameters	$\lambda \in (0, \infty)$
Support	$x \in \mathbb{N} \cup \{0\}$
PMF	$\mathbb{P}(X = x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$
Mean	λ
Variance	λ

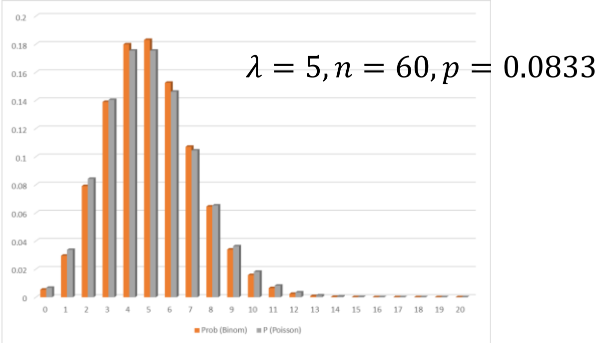
$\sqrt{\lambda}$

pmf



(with $n \to \infty, p \to 0$ and $np \to \lambda$, Poisson and Binomial converge)





Questions

PEAKS2TAILS

Exercise 3 – Show that the Poisson pmf is a proper pmf

Exercise 4 – An insurance sales-person sells 3 life insurance policies per week on an average. Use Poisson's law to calculate the probability that in a given month (4 weeks) he will sell

a. Some policies

b. 2 or more policies but less than 5 policies.

c. Assuming that there are 5 working days per week, what is the probability that in a given day he will sell one policy?

$$\lambda = 3 \ /\text{week} \rightarrow \lambda = 12 \ /\text{m.}$$

a) $P(X > 0) = 1 - P(X = 0) = 1 - \dfrac{e^{-12} \cdot (12)^0}{0!} = 1 - e^{-12}$

b) $P(2 \le X < 5) = P(X=2) + P(X=3) + P(X=4)$

$$= \frac{e^{-12} \cdot 12^2}{2!} + \frac{e^{-12} \cdot 12^3}{3!} + \frac{e^{-12} \cdot 12^4}{4!}$$

c) $\lambda = 3/5 \ /\text{day.}$

$$P(X=1) = \frac{e^{-3/5} \cdot (3/5)^1}{1!}$$

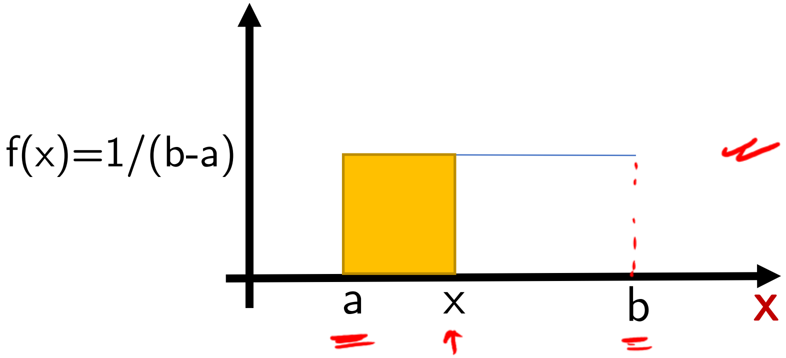
(continuous distributions)

Uniform Distribution

- Mostly used to draw random numbers uniformly between an interval [0,1] for simulation purpose
- A uniform distribution has bounds [a, b], but a standard uniform has bounds [0,1]

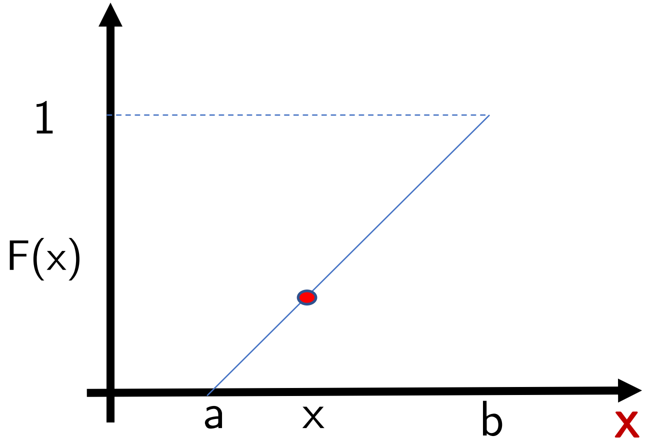
Parameters	$-\infty < a < b < \infty$
Support	$x \in [a, b]$
PDF	$\begin{cases} \frac{1}{b-a}, & x \in [a,b] \\ 0, & \text{otherwise} \end{cases}$
CDF	$\begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & x \in [a,b] \\ 1, & x > b \end{cases}$
Mean	$\frac{a+b}{2}$
Variance	$\frac{(b-a)^2}{12}$

$\frac{n^2-1}{12}$



$(b-a) \times c = 1$

$c = \frac{1}{b-a}$



Questions

Exercise 5 – For a random variable $X \sim U(0{,}23)$, find

a) $\mathbb{P}(2 < X < 18)$

b) $\mathbb{P}(X > 12 \mid X > 8)$

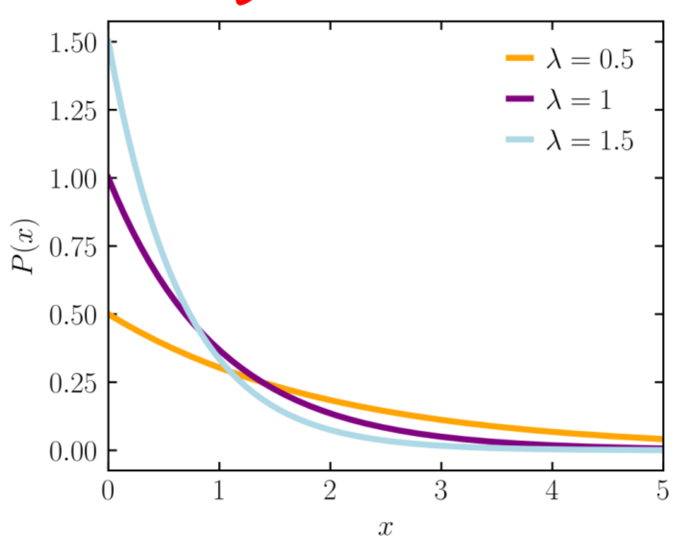
Exercise 6 – If U is standard uniform, prove that $aU + b$ is also uniform $(a, a+b)$. Using this property, what would be the distribution of $2U - 1$?

Exponential Distribution

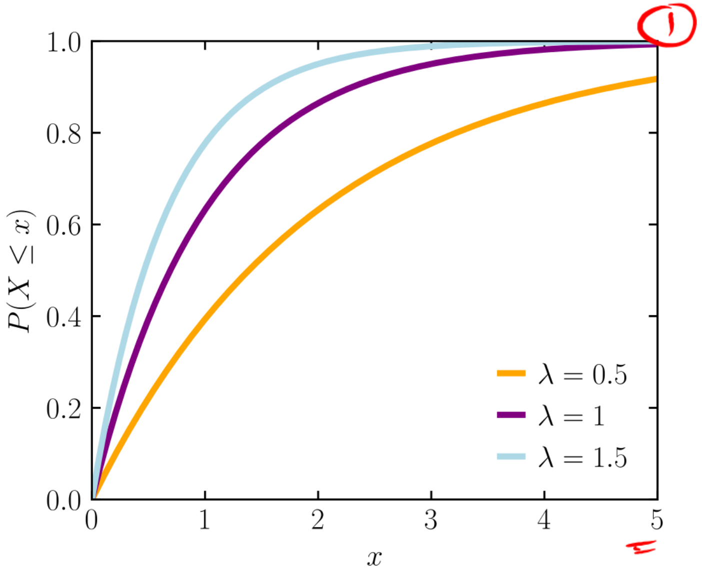
- Used to model time to failure/default. It's the basis of survival analysis
- If arrival is Poisson(λ) , then the time elapsed between successive arrival is Exponential(λ)

Parameters	$\lambda > 0$
Support	$x \in [0, \infty)$
PDF	$\lambda e^{-\lambda x}$
CDF	$1 - e^{-\lambda x}$
Mean	$\boxed{1/\lambda}$
Variance	$\boxed{1/\lambda^2}$

pdf



cdf



Questions

Exercise 6 – Let the hazard rate for a corporate bond be $\lambda = 2\%$ per year. Find the following

a) Probability of survival till 2 years

b) Probability of default before 5 years

c) Probability of default within 2 years to 5 years

d) Probability of surviving an additional 2 years after surviving for 3 years

a) $\mathbb{P}(\tau > 2) = e^{-2\% \times 2}$

b) $\mathbb{P}(\tau < 5) = 1 - e^{-2\% \times 5}$

c) $\mathbb{P}(2 < \tau < 5) = F(5) - F(2) = \left(1 - e^{-2\% \times 5}\right) - \left(1 - e^{-2\% \times 2}\right)$

$= e^{-2\% \times 2} - e^{-2\% \times 5}$

d) $\mathbb{P}(\tau > 5 \mid \tau > 3) = \dfrac{\mathbb{P}(\tau > 5, \tau > 3)}{\mathbb{P}(\tau > 3)} = \dfrac{\mathbb{P}(\tau > 5)}{\mathbb{P}(\tau > 3)} = \dfrac{e^{-2\% \times 5}}{e^{-2\% \times 3}} = e^{-2\% (5-3)}$

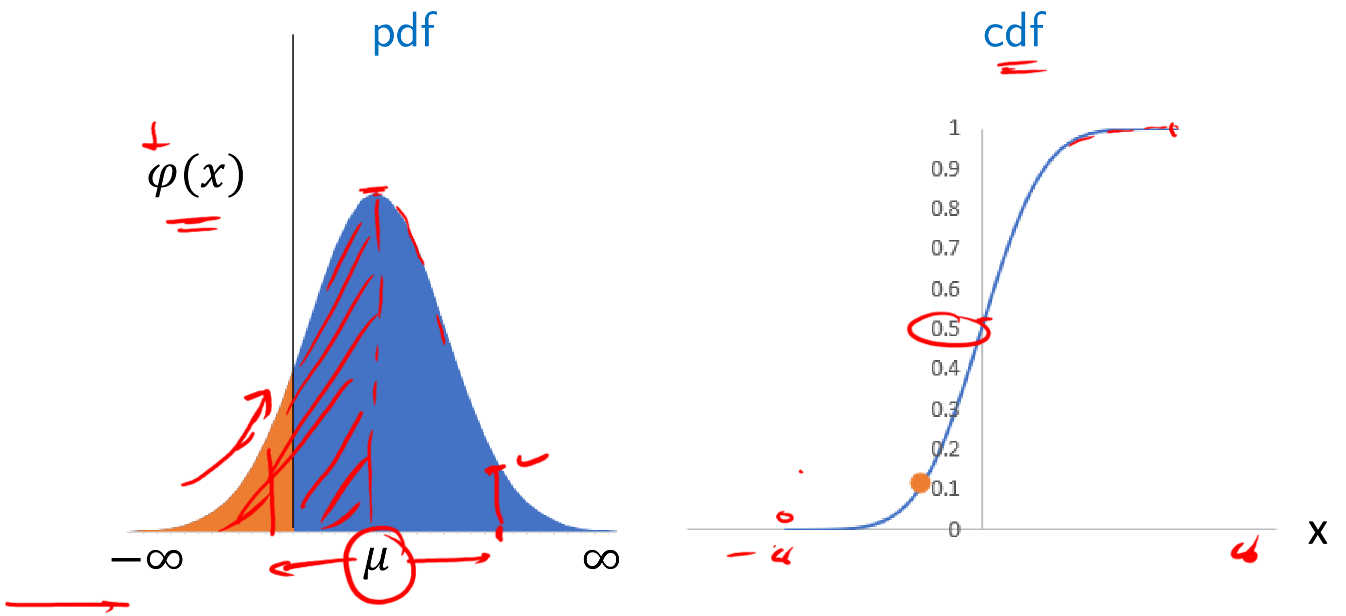
Normal Distribution

- The mother of all distributions. We model asset returns as normal.
- Normal distribution is important because of CLT

$f(x)$ ϕ

$X \sim \mathcal{N}(\mu, \sigma^2)$

Parameters	μ, σ^2
Support	$x \in \mathbb{R}$ $(-\infty, \infty)$
PDF	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
CDF	(no closed form formula)
Mean	μ
Variance	σ^2



Normal Distribution vs Standard Normal

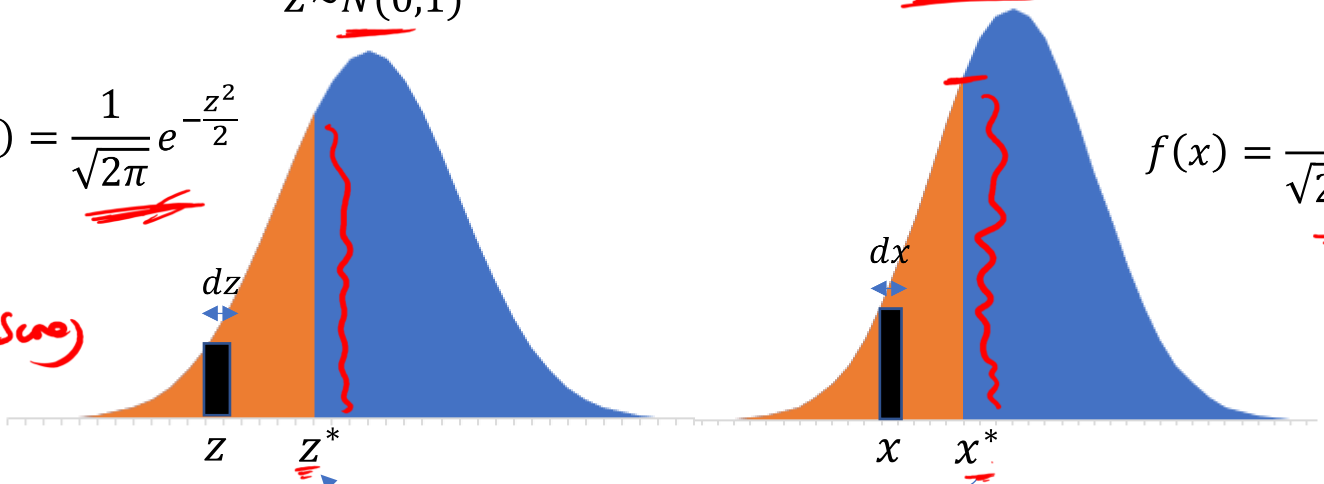
- All probability calculations are done under standard normal distribution
- The densities are proportional at a given point, however the interval probabilities are the same

$$X = \mu + \sigma Z$$

$Z \sim N(0,1)$ $\qquad$ $X \sim N(\mu, \sigma^2)$

$$g(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

(z-score)



$$Z = \frac{X - \mu}{\sigma}$$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f(x)dx = g(z)dz$$

$$\int_{-\infty}^{z} g(z)dz = \int_{-\infty}^{x} f(x)dx$$

$$P(Z \leq z) = P(X \leq x)$$

Questions

Exercise 7 – The critical values of the standard normal distribution are given as follows $P(Z \geq 1.645) = 5\%$, $P(Z \geq 1.96) = 2.5\%$ and $P(Z \geq 2.33) = 1\%$. Find the following probabilities

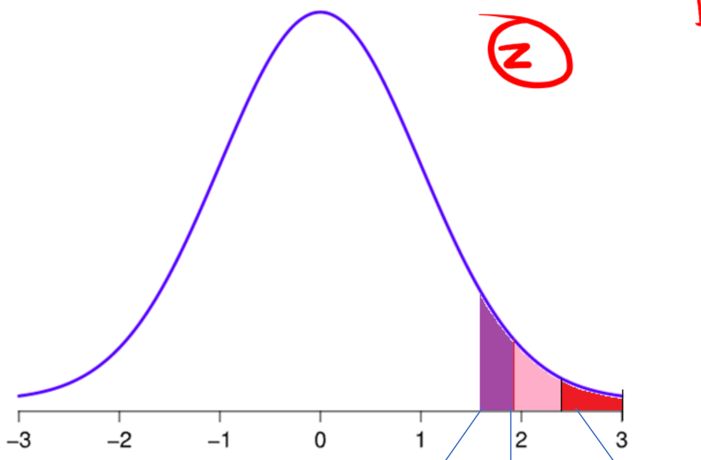
a) $P(Z \geq -1.645)$

b) $P(-1.96 \leq Z \leq 1.645)$

$= F(1.645) - F(-1.96)$

$= 95\% - 2.5\%$

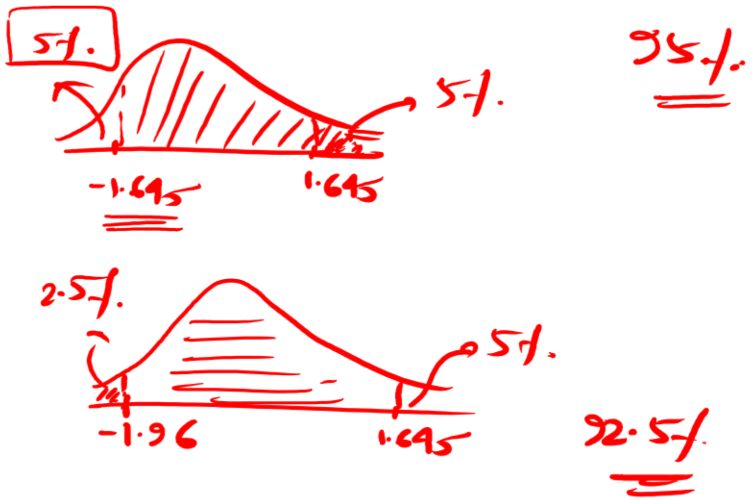
$= 92.5\%$



$P(Z \geq 1.645) = 5\%$

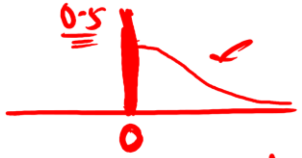
$P(Z \geq 2.33) = 1\%$

$P(Z \geq 1.96) = 2.5\%$



Questions

Exercise 8 – Let Z_1 and Z_2 be 2 standard normal random variables. Given the property that the linear combination of 2 normal randoms is a normal random, find $\mathbb{P}(Z_1 > Z_2)$. Does the answer depend on the correlation of Z_1 and Z_2 ?



Exercise 9 – If Z is a standard normal, find $\mathbb{E}[\max(Z, 0)]$ and $\mathbb{E}[Z \mid Z > 0]$

$\int_0^\infty z f(z)\,dz$, $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$

$= \int_0^\infty z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$, $\frac{1}{2}z^2 = y$, $z\,dz = dy$

$= \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-y} dy = \frac{1}{\sqrt{2\pi}}\left[e^{-y}\right]_\infty^0 = \frac{1}{\sqrt{2\pi}}$

$\mathbb{E}[\max(Z,0)]$

$= \int_{-\infty}^{\infty} \max(z,0) \cdot f(z)\,dz$

$= \int_0^\infty z f(z)\,dz$

$= \frac{1}{\sqrt{2\pi}}$

$\int_0^\infty z \left(\frac{f(z)}{\mathbb{P}(Z>0)}\right) dz$

$= \frac{2}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}}$

Lognormal Distribution

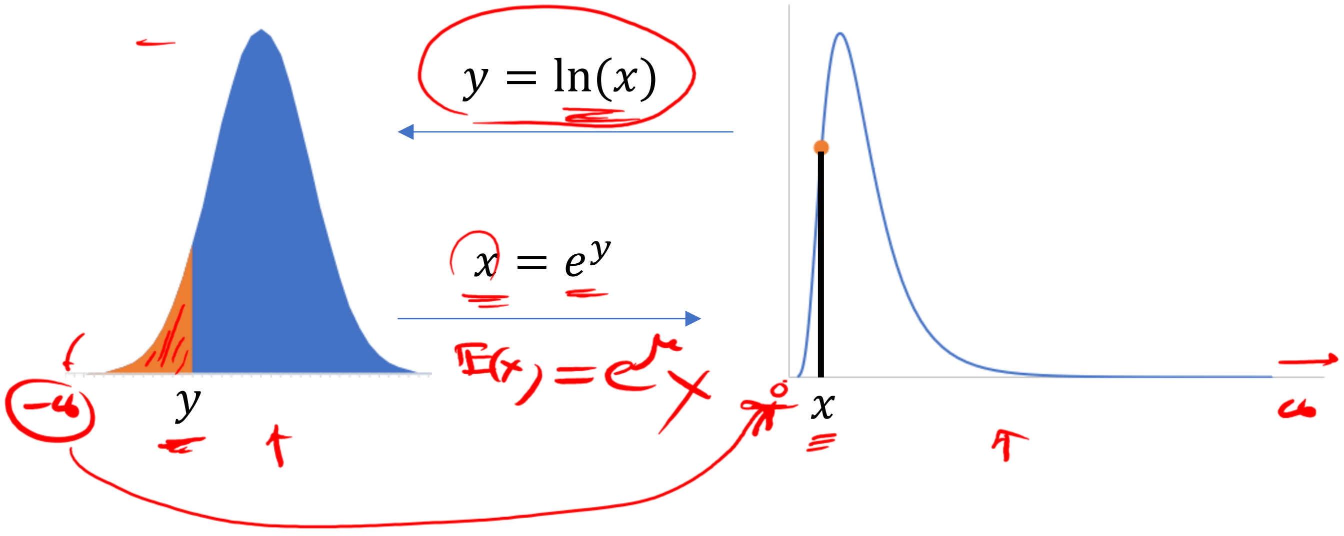
PEAKS2TAILS

- If $Y \sim N(\mu, \sigma^2)$, then $X = e^Y \sim \text{lognorm}(\mu, \sigma^2)$
- Similarly, if X is lognormal then $\ln X$ is normal
- We use lognormal distribution to model asset prices

$g(y).dy = f(x).dx$

$f(x) = g(y)\left(\frac{dy}{dx}\right) = \frac{1}{x} g(y)$

$Y \sim N(\mu, \sigma^2)$ $\quad$ $X \sim \text{lognormal}(\mu, \sigma^2)$



Parameters	μ, σ^2
Support	$x \in \mathbb{R}$
PDF	$\frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$
CDF	(no closed form formula)
Mean	$e^{\mu + \frac{1}{2}\sigma^2}$
Variance	$(e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$

Questions

PEAKS2TAILS

Exercise 10 – Define daily log return of a stock price as

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

If $r_t \sim N(0.1\%, 5\%)$, and current stock price = \$ 100, what is the expected stock price tomorrow ?

$$\left(\frac{S_t}{S_{t-1}}\right) = e^{r_t}$$

$$E\left[\frac{S_t}{S_{t-1}}\right] = e^{(0.1\% + \frac{1}{2} 5\%)}$$

$$E[S_t] = (\$100) \cdot e^{(0.1\% + \frac{1}{2} 5\%)}$$